VTPEH6270_CP04

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```r
setwd("C:/Users/Clair/Documents/VTPEH6270")

# Load file
School_immun_data<-read.csv("school_immun_2012_2019.csv")

# Function to convert percentage strings to numeric values
pct_to_num <- function(x) {
  x <- gsub("%", "", x)
  x[x == ""] <- NA
  as.numeric(x)
}

# Subset data to the 2018-2019 school year
immun_1819 <- School_immun_data[School_immun_data$Report.Period == "2018-2019", ]

# Keep only Public and Private schools
immun_1819 <- immun_1819[immun_1819$Type %in% c("Public School", "Private School"), ]
```

1 Objective and Approach

1.1 Specific Objective

This research question examines whether the proportion of religious vaccination exemptions differs between public and private schools.

1.2 Data Description

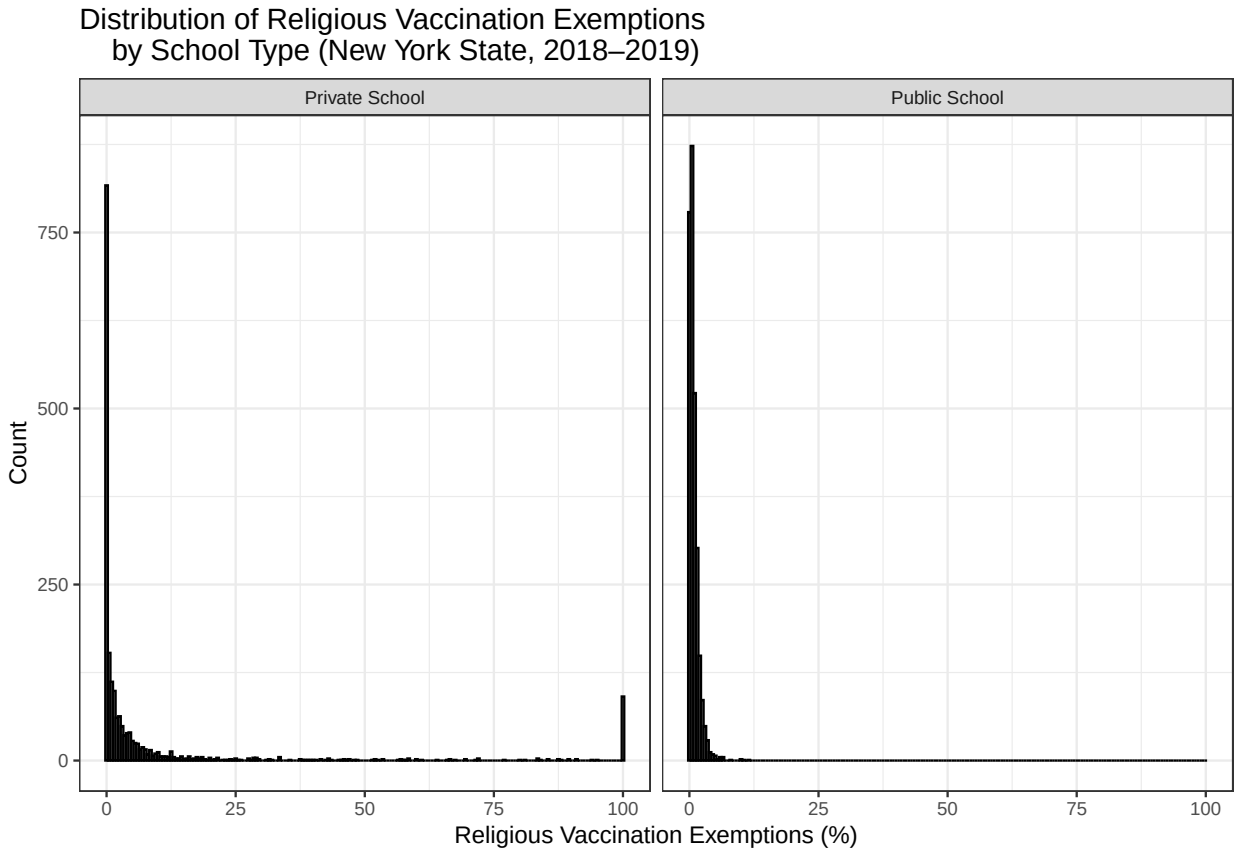
```r
library(knitr)

table_variables <- data.frame(
  Variable_Name = c("Type", "Percent Religious Exemptions"),
  Variable_Type = c("Categorical", "Continuous"),
  Variable_Class = c("character", "numeric")
)

kable(table_variables, caption = "Description of variables used in the analysis")
```

Table 1: Description of variables used in the analysis

Variable_Name	Variable_Type	Variable_Class
Type	Categorical	character
Percent Religious Exemptions	Continuous	numeric

1.3 Data Visualization

```r
#  Convert Religious Vaccination Exemptions from percentage to numeric
immun_1819$Percent.Religious.Exemptions <- pct_to_num(immun_1819$Percent.Religious.Exemptions)

library(ggplot2)
ggplot(immun_1819, aes(x = Percent.Religious.Exemptions)) +
  geom_histogram(binwidth = 0.5, fill = "grey70", color = "black") +
  facet_wrap(~Type) +
  labs(
    title = "Distribution of Religious Vaccination Exemptions
    by School Type (New York State, 2018-2019)",
    x = "Religious Vaccination Exemptions (%)",
    y = "Count"
  ) +
  theme_bw(base_size = 9)
```



1.4 Description of Plausible Relationships

Visual inspection of the histograms suggests that religious vaccination exemptions differ by school type. Most schools show exemption rates close to zero, but private schools appear to have greater variability and more extreme values. This suggests that higher exemption rates may occur more frequently in private schools than in public schools.

1.5 Parameters of Interest

```
library(knitr)

parameter_table <- data.frame(
  Parameter = c(
    "Mean religious exemption rate in public schools",
    "Mean religious exemption rate in private schools",
    "Difference in mean exemption rates"
  ),
  Symbol = c("mu[1]", "mu[2]", "Delta = mu[2] - mu[1]"),
  Description = c(
    "Average percentage of religious vaccination exemptions among public schools (%)",
    "Average percentage of religious vaccination exemptions among private schools (%)",
    "Effect size: difference in average religious exemption
    percentage between private and public schools (%)"
```

```
  )
)

kable(parameter_table, caption = "Parameters of interest for the association
      between school type and religious vaccination exemptions")
```

Table 2: Parameters of interest for the association between school type and religious vaccination exemptions

Parameter	Symbol	Description
Mean religious exemption rate in public schools	mu[1]	Average percentage of religious vaccination exemptions among public schools (%)
Mean religious exemption rate in private schools	mu[2]	Average percentage of religious vaccination exemptions among private schools (%)
Difference in mean exemption rates	Delta = mu[2] - mu[1]	Effect size: difference in average religious exemption percentage between private and public schools (%)

2 Simulation

2.1 Simulation Basis

```
# Set seed for reproducibility
set.seed(123)

# Number of observations
nn <- 2000

# Probability of each school type
prob_private <- 0.25

# Simulate school type
data_sim <- data.frame(
  Type = sample(
    c("Public School", "Private School"),
    nn,
    replace = TRUE,
    prob = c(1 - prob_private, prob_private)
  )
)

# Parameters for religious exemption rate
mean_public <- 0.5
mean_private <- 2.0
sd_public <- 1.0
sd_private <- 5.0
```

```r
# Create empty outcome variable
data_sim$Percent.Religious.Exemptions <- NA

# Simulate exemption rate conditional on school type
for (ii in 1:nrow(data_sim)) {
  if (data_sim$Type[ii] == "Public School") {
    data_sim$Percent.Religious.Exemptions[ii] <- rnorm(
      1,
      mean = mean_public,
      sd = sd_public
    )
  } else if (data_sim$Type[ii] == "Private School") {
    data_sim$Percent.Religious.Exemptions[ii] <- rnorm(
      1,
      mean = mean_private,
      sd = sd_private
    )
  }
}

data_sim$Percent.Religious.Exemptions <-
  pmax(data_sim$Percent.Religious.Exemptions, 0)

data_sim$Percent.Religious.Exemptions <-
  pmin(data_sim$Percent.Religious.Exemptions, 100)

head(data_sim)
```

2.2 Simulation Output

```r
summary_sim <- aggregate(
  Percent.Religious.Exemptions ~ Type,
  data = data_sim,
  FUN = function(x) c(
    Mean = mean(x),
    SD = sd(x)
  )
)

print(summary_sim)
```

```
##             Type Percent.Religious.Exemptions.Mean
## 1 Private School                         3.1216811
## 2  Public School                         0.7071537
##   Percent.Religious.Exemptions.SD
## 1                       3.6185928
## 2                       0.7424722
```

2.3 Simulation Function

```r
simulate_exemptions <- function(effect_size, noise, sample_size) {

mean_public <- 0.5
mean_private <- mean_public + effect_size

  # simulate school type
  Type <- sample(
    c("Public School", "Private School"),
    sample_size,
    replace = TRUE,
    prob = c(0.75, 0.25)
  )

  # simulate exemption rate
  exemptions <- numeric(sample_size)

  for (i in 1:sample_size) {

    if (Type[i] == "Public School") {
      exemptions[i] <- rnorm(1, mean_public, noise)
    } else {
      exemptions[i] <- rnorm(1, mean_private, noise)
    }

  }

  exemptions <- pmin(pmax(exemptions, 0), 100)

  # Combine data
  data_sim <- data.frame(
    Type = Type,
    Percent.Religious.Exemptions = exemptions
  )

summary_sim <- aggregate(
  Percent.Religious.Exemptions ~ Type,
  data = data_sim,
  FUN = function(x) c(
    Mean = mean(x),
    SD = sd(x)
  )
)

return(list(
  data = data_sim,
  summary = summary_sim
))

}
```

2.4 Simulation Automation

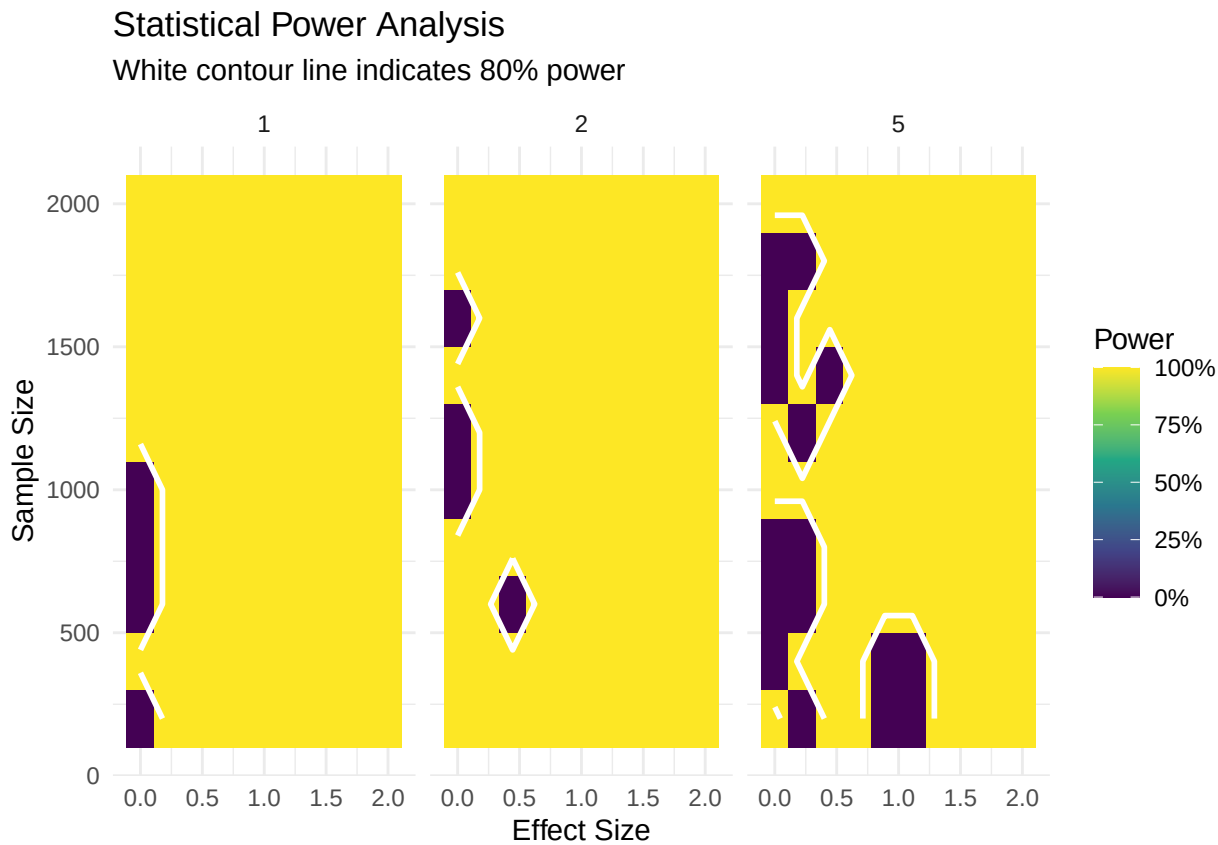
```
effect_sizes <- seq(0, 2, length.out = 10)

sample_sizes <- seq(200, 2000, length.out = 10)

noise_levels <- c(1, 2, 5)

simulation_results <- list()

counter <- 1

for (e in effect_sizes) {
  for (n in sample_sizes) {
    for (noise in noise_levels) {

      result <- simulate_exemptions(
        effect_size = e,
        noise = noise,
        sample_size = n
      )

      simulation_results[[counter]] <- list(
        effect_size = e,
        sample_size = n,
        noise = noise,
        data = result$data,
        summary = result$summary
      )

      counter <- counter + 1

    }
  }
}
```

3 Visualization

```
results_df <- data.frame()

for (i in 1:length(simulation_results)) {

  sim <- simulation_results[[i]]

  summary_sim <- sim$summary

  mean_public <- summary_sim$Percent.Religious.Exemptions[
    summary_sim$Type == "Public School","Mean"
  ]
```

```r
  mean_private <- summary_sim$Percent.Religious.Exemptions[
    summary_sim$Type == "Private School","Mean"
  ]

  observed_effect <- mean_private - mean_public

  results_df <- rbind(
    results_df,
    data.frame(
      effect_size = sim$effect_size,
      sample_size = sim$sample_size,
      noise = sim$noise,
      observed_effect = observed_effect
    )
  )
}

results_df$detected <- results_df$observed_effect > 0

power_df <- aggregate(
  detected ~ effect_size + sample_size + noise,
  data = results_df,
  mean
)

colnames(power_df)[4] <- "power"

ggplot(power_df,
             aes(x = effect_size,
                 y = sample_size,
                 fill = power)) +

  geom_tile() +

  scale_fill_viridis_c(
    name = "Power",
    labels = scales::percent_format()
  ) +

  geom_contour(
    aes(z = power),
    breaks = 0.8,
    color = "white",
    linewidth = 1
  ) +

  facet_wrap(~noise) +

  labs(
    x = "Effect Size",
    y = "Sample Size",
    title = "Statistical Power Analysis",
    subtitle = "White contour line indicates 80% power"
```

```
  ) +

  theme_minimal()
```



4 AI Use Disclosure Statement

This document was completed with assistance from ChatGPT. AI was used to help recall R functions and to provide explanations and examples of their usage. All code and interpretations were reviewed and finalized by the author.